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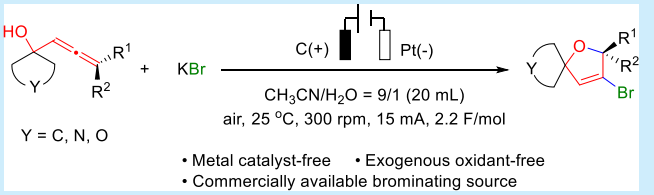
Electrochemical Cyclization of 2,3-Allenols

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ABSTRACT: An efficient electrochemical bromocyclization of allenols has been realized for the synthesis of spirocyclic 2,5-dihydrofurans. The reaction used commercially available and nontoxic KBr as the brominating source in a simple setup under open-air conditions. Notably, optically active products can be obtained from optically active 2,3-allenols without any racemization, further enhancing the synthetic utility.



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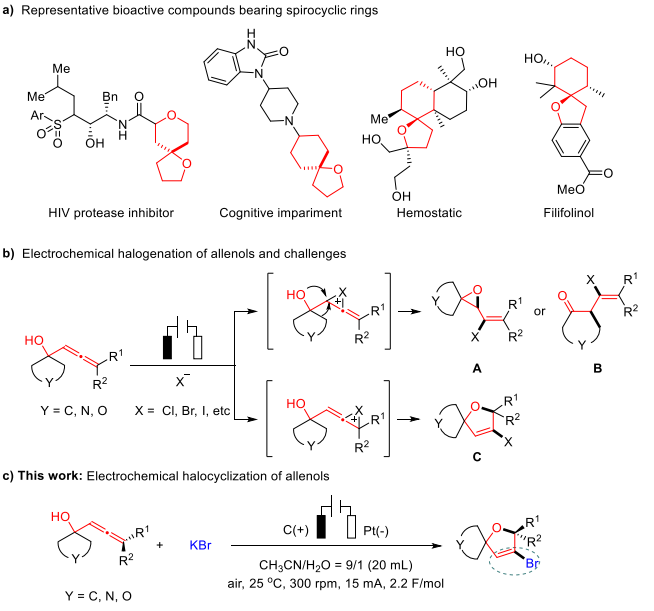
Spirocyclic frameworks are prevalent in natural products or pesticides and frequently recognized as chiral scaffolds for synthesizing chiral ligands.[1−3] Of particular interest are the oxygen-containing spirocyclic compounds with their enhanced three-dimensional structures, which are significant motifs in various drugs and biologically active molecules, such as hemostatic and filifolinol, due to their interactions with specific protein receptors or enzymes (Scheme 1a).[4−6] Thus, the wide-range importance of these compounds has driven the advancement of new synthetic strategies to facilitate their production.[7]

In recent years, electrochemical anodic oxidations have emerged as mild, efficient, and environmentally friendly strategies for the synthesis of complex molecules.[8] Notably, the electrochemical halogenation of alkenes or alkynes using commercially available, nontoxic, and atom-efficient MX/HX systems has been developed to avoid the use of hazardous or expensive halogenation reagents (such as dihalogen or *N*-halogensuccinimide), transition metal salts (e.g., CuX_2, FeX_3, etc.), and stoichiometric strong oxidants.[9−12]

Based on our previous reports on the easy synthesis of tertiary allenols,[13] we investigated whether the electrochemical halocyclization of allenols[14,15] could serve as an efficient and convenient method for accessing oxygen-containing spirocyclic products **C**. However, two challenges remained (Scheme 1b): first, engaging different double bonds with similar properties in allenes could yield spirocyclic products **A**;[16] second, ring-expansion reactions could produce nonspirocyclic products **B**.[12a] Herein, we describe our efforts to develop the electrochemical bromocyclization of allenols for the synthesis of spirodihydrofuran with a useful C−Br bond handle for downstream functionalizations as a starting point (Scheme 1c).

To assess the feasibility of this strategy, we commenced our study by reacting allenol **1a** with KBr. Gratifyingly, the spirocyclic product **2a** was obtained in 87% yield using a graphite plate as the anode and a platinum plate as the cathode, with CH_3CN/H_2O (9/1) as the solvent under a constant current of 15 mA in an undivided cell (Table 1, entry 1). Control experiment confirmed that electricity was essential (Table 1, entry 2). Other bromide salts were investigated, with NaBr exhibiting a similar efficiency to KBr, while $NiBr_2$ demonstrated a lower efficiency (Table 1, entries 3 and 4). Decreasing or increasing the equivalents of KBr resulted in slightly lower yields (Table 1, entries 5 and 6). Screening the solvent ratio of CH_3CN/H_2O revealed that a 9:1 ratio was

Scheme 1. Electrochemical Halofunctionalization of Allenols



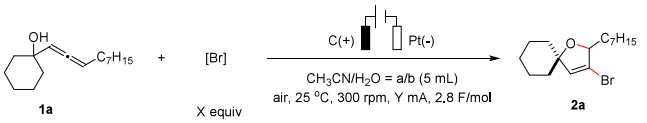
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Table 1. Optimization of the Reaction Conditions[a]



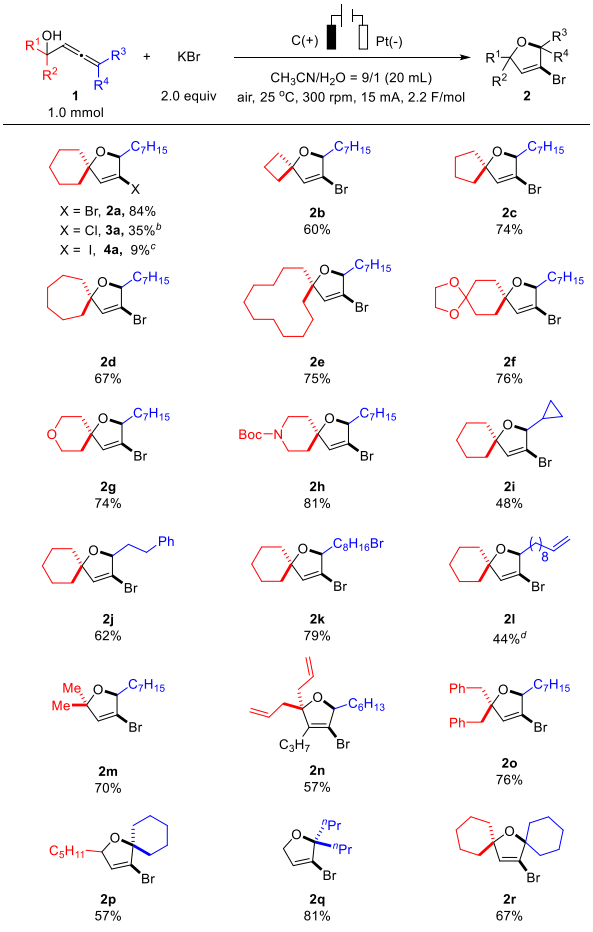
entry	[Br]	*a*/*b*	*X*	*Y*	*t*/h	yield[b] (%)
1	KBr	9/1	2.0	15	1	87
2[c]	KBr	9/1	2.0	0	1	0
3	NaBr	9/1	2.0	15	1	86
4	$NiBr_2$	9/1	1.0	15	1	74
5	KBr	9/1	1.5	15	1	75
6	KBr	9/1	2.5	15	1	73
7	KBr	4/1	2.0	15	1	64
8	KBr	19/1	2.0	15	1	62
9	KBr	9/1	2.0	3	5	69
10	KBr	9/1	2.0	5	3	83
11	KBr	9/1	2.0	8	2	85
12[d]	KBr	9/1	2.0	30	0.5	71
13[e]	KBr	9/1	2.0	15	5	76
14[f]	KBr	9/1	2.0	15	4	86 (84)[g]

[a]Reaction conditions unless specified otherwise: **1a** (0.2 mmol), CH_3CN/H_2O (5 mL), graphite plate anode (10 × 10 × 3 mm), Pt plate cathode (10 × 10 × 0.1 mm), undivided cell, air, 25 °C, 300 rpm, 2.8 F/mol. [b]The NMR yield of **2a** was determined by ^1H NMR analysis using dibromomethane as the internal standard. [c]No current. [d]3.0 F/mol. [e]**1a** (1.0 mmol) in the solvent CH_3CN/H_2O (22.5/2.5 mL). [f]**1a** (1.0 mmol) in the solvent CH_3CN/H_2O (18.0/2.0 mL), 2.2 F/mol. [g]Isolated yield.

optimal (Table 1, entries 7 and 8 vs entry 1). Further examination showed that the yield of **2a** decreased with either a lower or higher current (Table 1, entries 9−12). Scaling up the reaction to 1.0 mmol resulted in a moderate yield of 76% (Table 1, entry 13). Through further efforts, we found that increasing the reaction concentration, decreasing the reaction time could enhance the production of **2a**, which was isolated in 84% yield on the 1.0 mmol scale (Table 1, entry 14).

With the optimized reaction conditions established, we explored the substrate scope of this electrochemical bromocyclization protocol of allenols (Scheme 2). As a comparison, the reaction of **1a** with KCl and KI under the standard conditions afforded 3-chloro- and 3-iodo-dihydrofuran products **3a** and **4a** in 35% and 9% yields, respectively. A wide range of tertiary 2,3-allenols underwent bromocyclization smoothly, yielding 2,5-dihydrofuran derivatives **2** in good to excellent yields. First, we evaluated the ring sizes of cyclic tertiary 2,3-allenols. Various substrates with carbocycles of different sizes proved viable in our electrochemical transformation, generating a range of spirocyclic products in yields of 60−84%. Notably, highly strained four-membered allenol **1b** and medium-sized 12-membered allenol **1e** served as suitable substrates, affording products **2b** and **2e** in 60% and 75% yields, respectively. An acetal moiety may also survive, yielding 76% of **2f**. Moreover, substrates containing heterocycles performed well, providing the corresponding oxa- or aza-spirocyclic products **2g** and **2h** in yields of 74% and 81%, respectively. A variety of synthetically useful groups, including cyclopropyl (**2i**), phenyl (**2j**), halide (**2k**), and allyl (**2l**) groups were well-tolerated. Beyond cyclic tertiary allenols, this electrochemical strategy may also be extended to acyclic tertiary allenols [dimethyl (**1m**), diallyl (**1n**), and dibenzyl (**1o**)]. Remarkably, a trisubstituted allenol enabled access to a

Scheme 2. Substrate Scope Study[a]



[a]Reaction conditions unless specified otherwise: **1a** (1.0 mmol), KBr (2.0 mmol), CH_3CN/H_2O = 9/1 (20 mL), graphite plate anode (10 × 10 × 3 mm), Pt plate cathode (10 × 10 × 0.1 mm), undivided cell, constant current 15 mA, air, 25 °C, 300 rpm, 2.2 F/mol. [b]KCl (2.0 mmol) was used instead of KBr (2.0 mmol). [c]KI (2.0 mmol) was used instead of KBr (2.0 mmol). [d]2.8 F/mol (see more details in the Supporting Information).

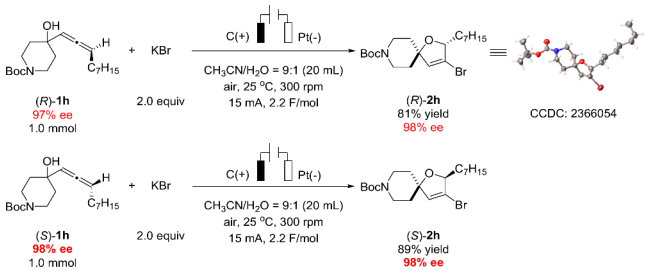
4-bromide-substituted spirocyclic 2,5-dihydrofuran product **2p** in 57% yield. The protocol also proved effective for primary allenol **1q**, where both R^3 and R^4 are n-propyl. Finally, bispirocyclic product **2r** was smoothly synthesized.

Furthermore, when optically active allenols (*R*)-**1h** and (*S*)-**1h** were subjected to the electrochemical bromocyclization protocol under the standard conditions, the corresponding chiral spirocyclic products (*S*)-**2h** and (*R*)-**2h** were obtained efficiently without any loss of *ee*. A single-crystal X-ray diffraction analysis was conducted to determine the absolute configuration of (*R*)-**2h** (Scheme 3).

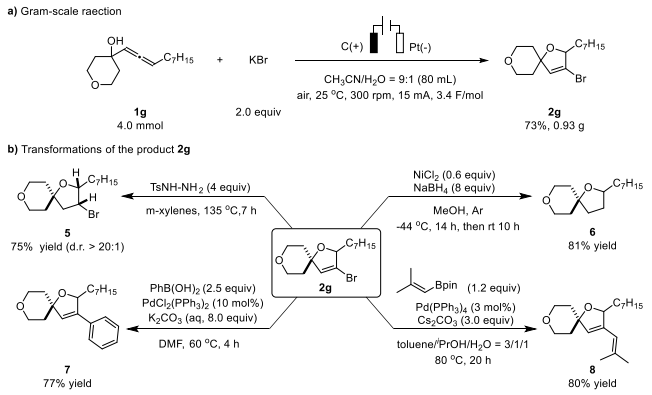
The reaction was successfully scaled up to a 4.0 mmol scale using a simple and readily made device (see Figure S3 in the Supporting Information), yielding product **2g** in 73% yield (Scheme 4a). The synthetic utility of the spirocyclic products is demonstrated (Scheme 4b): Product **2g** was readily reduced by *p*-toluenesulfonyl hydrazide at 135 °C to afford spiro-tetrahydrofuranyl bromide **5** in a highly stereoselective manner.[17] By comparison, when **2g** was subjected to 0.6 equiv of $NiCl_2$ and 8 equiv of $NaBH_4$, the debrominated spiro-tetrahydrofuran **6** was obtained in 81% yield.[18] Phenyl[19] and alkenyl[20] groups were introduced into the spirocyclic 2,5-

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Scheme 3. Chirality Transfer in Electrochemical Halocyclization of *a*-Allenols



Scheme 4. Gram-Scale Reaction and Transformations of the Products



dihydrofuran derivatives via Suzuki-Miyaura coupling, yielding compounds **7** and **8** in 77% and 80% yields, respectively.

The reaction was monitored by cyclic voltammetry (Figure 1). Two oxidation peaks of KBr were observed at 1.0 and 1.4 V

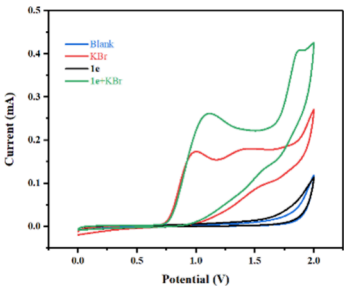
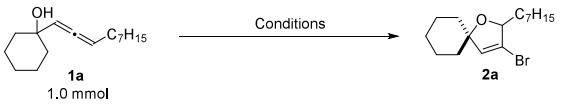


Figure 1. Cyclic voltammograms of background (blue line), KBr (0.08 mmol) (red line), **1e** (0.04 mmol) (black line), and KBr (0.08 mmol) and **1e** (0.04 mmol) (green line) in $MeCN/H_2O$ (4.5/0.5 mL) with $LiClO_4$ (0.2 M) under air atmosphere. Scan rate: 100 mV/s.

(vs SCE),[21] corresponding to the anodic oxidation of Br^- to the bromine radical, which subsequently dimerized to form molecular Br_2 and combined with Br^- to produce Br_3^-. The second oxidative peak corresponded to the oxidation of Br_3^- to Br_2. When KBr and allenol **1e** were combined, an increase in the oxidative peak current was noted, indicating that the chemical reaction occurs between electrochemically generated molecular Br_2 and allenol **1e** (Figure 1, green curve).

Control experiments showed that when substrate **1a** was reacted with molecular Br_2—either added directly or dropwise—instead of KBr and electricity, product **2a** was obtained, but with significantly reduced yields of 28% and 25%, respectively (Scheme 5, entries 2 and 3), indicating the advantage of the electrosynthesis. Interestingly, the addition of

Scheme 5. Electrosynthesis vs Normal Bromocyclization

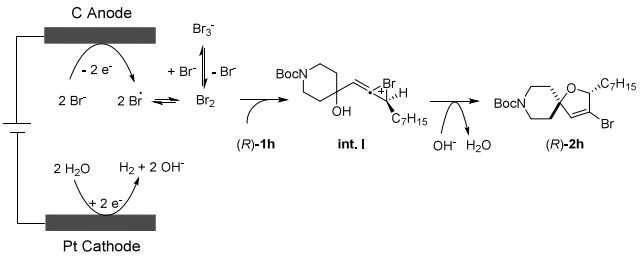


Entry	Conditions	NMR yield of **2a**(%)
1	standard conditions (Table 1, entry 14)	86
2	Br_2 (2.0 equiv) CH_3CN (18 mL)/H_2O (2 mL), air, 25 °C, 4 h	28
3	Br_2 (2.0 equiv)/CH_3CN (4 mL) (dropwise over 1.5 h) CH_3CN (14 mL)/H_2O (2 mL), air, 25 °C, 2 h	25
4	Br_2 (2.0 equiv)/CH_3CN (4 mL) (dropwise over 1.5 h) KBr (2.0 equiv), CH_3CN (14 mL)/H_2O (2 mL), air, 25 °C, 2 h	47

KBr increased the yield of **2a** to 47% (Scheme 5, entry 4). We reasoned that the slow releasing of the Br_2 is responsible for the higher yield observed in the current study.

A plausible reaction mechanism is proposed in Scheme 6. At the anode, a bromide anion is oxidized to a bromine radical,

Scheme 6. Proposed Mechanism



subsequently combining to form molecular Br_2, which would interact with Br^- to produce Br_3^-. This Br_2 electrophilically interacted with (*R*)-**1h** to generate the bromonium ion intermediate **int. I**. Intramolecular *anti*-nucleophilic attack by the hydroxy group from the opposite side followed by deprotonation, yielded the final product (*R*)-**2h**. Concurrently, at the cathode, water is continuously reduced to generate OH^- and gaseous H_2.

In summary, we have described a highly efficient electrochemical synthetic method for spirocyclic 2,5-dihydrofuran derivatives from allenols. This approach operates under metal catalyst-free and exogenous oxidant-free conditions, utilizing commercially available and nontoxic KBr as the brominating source. The method enjoys ease of handling and mild reaction conditions preventing racemization with the optically active substrates. Further studies in this area including control of the cyclization mode are currently ongoing in our laboratory.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c00193.

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Experimental procedures, characterization data for obtained compounds, crystal data for (*R*)-**2h**, and NMR spectra (PDF)

Accession Codes

Deposition Number 2366054 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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